

Stress Invariants & Stress Deviation

Coefficients in the Characteristic Equation When we find out the principal stresses

$$I_1 = \tau_{xx} + \tau_{yy} + \tau_{zz}$$

$$I_2 = \tau_{xx}^2 + \tau_{yy}^2 + \tau_{zz}^2 - \tau_{xx}\tau_{yy} - \tau_{xx}\tau_{zz} - \tau_{yy}\tau_{zz}$$

$$I_3 = \tau_{xx}\tau_{yy}\tau_{zz} + 2\tau_{xy}\tau_{xz}\tau_{yz} - \tau_{xx}\tau_{yz}^2 - \tau_{yy}\tau_{xz}^2 - \tau_{zz}\tau_{xy}^2$$

These are independent of Co-ordinate System $\rightarrow$ Invariants.

In matrix terminology

$$I_1 = \text{trace}(\tau) \quad : \text{Sum of diagonal terms}$$

$$I_2 = \frac{1}{2} \left\{ \text{trace}(\tau^2) - [\text{trace}(\tau)]^2 \right\}$$

$$I_3 = \det(\tau)$$

In principal co-ordinate system

$$I_1 = \sigma_1 + \sigma_2 + \sigma_3$$

$$I_2 = -(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)$$

$$I_3 = \sigma_1\sigma_2\sigma_3$$

Few useful Relations

$$\tau_{xx}^2 + \tau_{yy}^2 + \tau_{zz}^2 + 2\tau_{xy}^2 + 2\tau_{xz}^2 + 2\tau_{yz}^2 = \sigma_1^2 + \sigma_2^2 + \sigma_3^2$$

Useful in study of stress strain laws & failure Criteria.

Octahedral stresses:- These are normal & shear tractions acting on planes having unit normal vectors in the principal stress Co-ordinate system

$$\text{Given by } \langle n_1, n_2, n_3 \rangle = \left\langle \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}} \right\rangle$$

Normal traction acting on any octahedral plane is

$$\sigma_{oct} = \sigma_1 n_1^2 + \sigma_2 n_2^2 + \sigma_3 n_3^2 = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3) = \frac{1}{3}I_1.$$

Magnitude of shear traction

$$|\tau_{oct}| = \frac{1}{3} \sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}$$

$$= \frac{\sqrt{2}}{3} \left\{ I_1^2 + 3I_2 \right\}^{1/2}$$

Hydrostatic and Deviatoric Stresses

Any stress matrix can be broken into two parts.

- Hydrostatic part
- Deviatoric part

$$\tau = \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{yz} & \tau_{zx} & \tau_{zz} \end{bmatrix} \quad \text{if } \tau_m \text{ is defined as}$$

$$\tau_m = \frac{1}{3}(\tau_{xx} + \tau_{yy} + \tau_{zz})$$

$$\tau = \underbrace{\begin{bmatrix} (\tau_{xx} - \tau_m) & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & (\tau_{yy} - \tau_m) & \tau_{yz} \\ \tau_{yz} & \tau_{zx} & (\tau_{zz} - \tau_m) \end{bmatrix}}_{\text{Deviatoric Part}} + \underbrace{\begin{bmatrix} \tau_m & 0 & 0 \\ 0 & \tau_m & 0 \\ 0 & 0 & \tau_m \end{bmatrix}}_{\substack{\text{Hydrostatic part} \\ \text{or} \\ \text{Isotropic part}}}$$

or, $\tau = S + \tau^{iso}$

or, $S = \tau - \tau^{iso}$

Sometimes S is denoted by τ^{dev}

$$\therefore \boxed{\tau^{dev} = \tau - \tau^{iso}}$$

Study of deformation behaviour.
In elastic range of deformations

- τ_m controls the Volumetric Change
- τ^{dev} controls the distortion.

According to Invariants of Stress deviators have been defined.